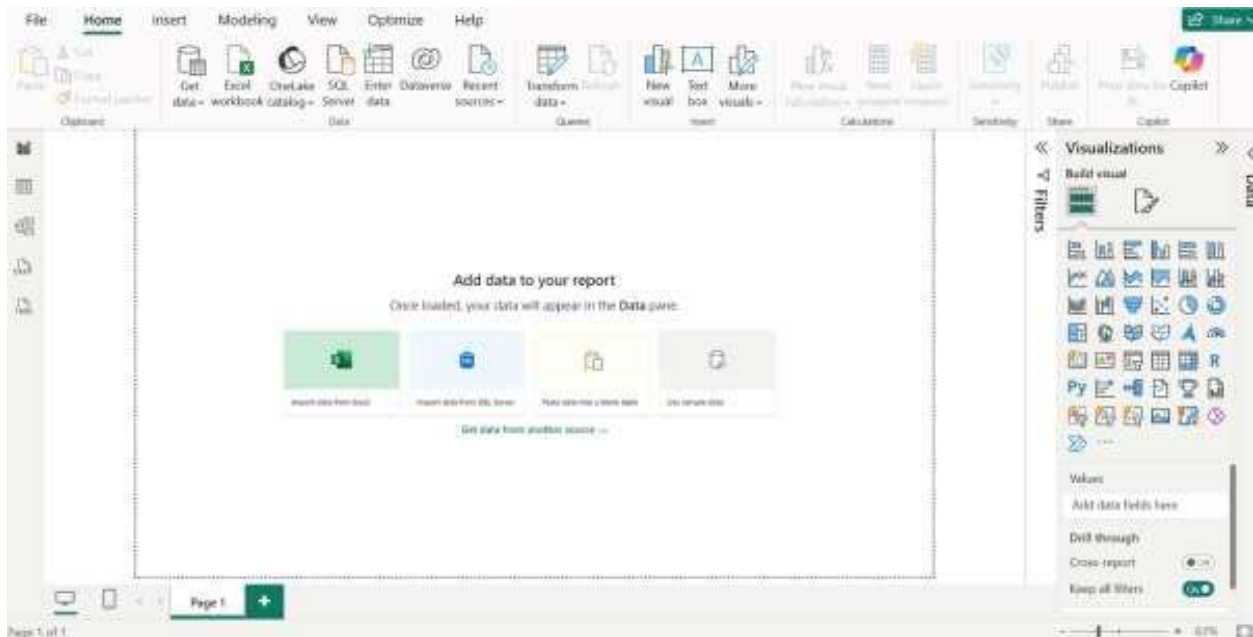


WEEK 1 – POWER BI DATA PREPARATION REPORT

Power BI Desktop – Starting the Analysis

The Power BI Desktop environment was used as the initial workspace for the report. The interface brings together tools for loading data, preparing tables, building relationships, and designing visual reports. This provides the foundation for carrying out the Superstore sales analysis.



Power BI Desktop interface used to begin the reporting task.

Importing and Loading the Superstore Dataset

The Superstore Sales CSV was selected as the source dataset. The import preview was checked to confirm that the fields and sample records were being recognized correctly. After verification, the data was loaded into Power BI so that it could be examined and transformed.

Home Insert Modeling View Optimize Help

Get data Excel OneLake data workbook catalog

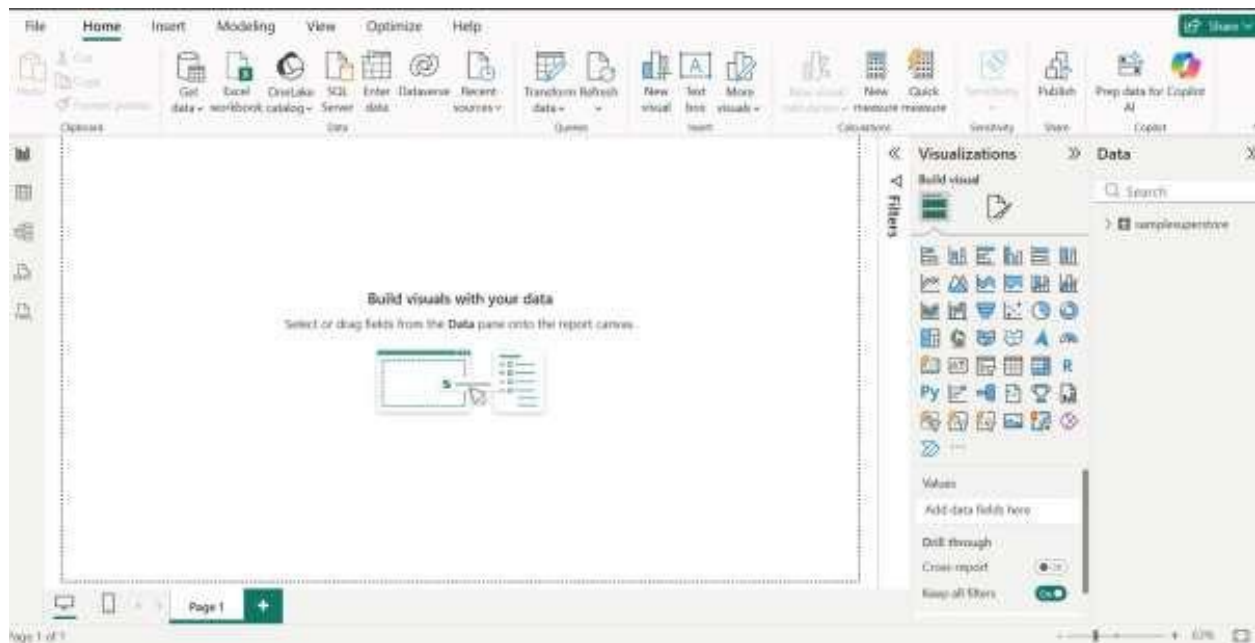
samplesuperstore (1).csv

File Origin: 65001:Unicode (UTF-8) Delimiter: Comma Data Type Detection: Based on first 200 rows

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City
1	US-2023-103000	1/1/2023	1/7/2023	Standard Class	DF-13000	Garen Powers	Consumer	United States	Phoenix
2	US-2023-112326	1/4/2023	1/6/2023	Standard Class	PO-18185	Polina Oler	Home Office	United States	Naperville
3	US-2023-112328	1/4/2023	1/8/2023	Standard Class	PO-18186	Polina Oler	Home Office	United States	Naperville
4	US-2023-112334	1/4/2023	1/8/2023	Standard Class	PO-18186	Polina Oler	Home Office	United States	Naperville
5	US-2023-141817	1/5/2023	1/12/2023	Standard Class	MD-18085	Mark Bryson	Consumer	United States	Philadelphia
6	US-2023-187189	1/4/2023	1/10/2023	Standard Class	ME-17320	Maria Elstad	Home Office	United States	Portland
7	US-2023-187199	1/4/2023	1/10/2023	Standard Class	ME-17320	Maria Elstad	Home Office	United States	Portland
8	US-2023-104034	1/4/2023	1/7/2023	First Class	JD-21145	Jack O'Brian	Corporate	United States	Athens
9	US-2023-187199	1/4/2023	1/10/2023	Standard Class	ME-17320	Maria Elstad	Home Office	United States	Portland
10	US-2023-187199	1/4/2023	1/10/2023	Standard Class	ME-17320	Maria Elstad	Home Office	United States	Portland
11	US-2023-187199	1/4/2023	1/10/2023	Standard Class	ME-17320	Maria Elstad	Home Office	United States	Portland
12	US-2023-130813	1/6/2023	1/6/2023	Second Class	LS-17230	Lisa Sanders	Consumer	United States	Lincoln
13	US-2023-187199	1/4/2023	1/10/2023	Standard Class	ME-17320	Maria Elstad	Home Office	United States	Portland
14	US-2023-187199	1/4/2023	1/10/2023	Standard Class	ME-17320	Maria Elstad	Home Office	United States	Portland
15	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivik Sanderson	Consumer	United States	Houston
16	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivik Sanderson	Consumer	United States	Houston
17	US-2023-125409	1/5/2023	1/13/2023	Standard Class	MS-17890	Melanie Seta	Consumer	United States	Laredo
18	US-2023-125409	1/5/2023	1/13/2023	Standard Class	MS-17890	Melanie Seta	Consumer	United States	Laredo
19	US-2023-149035	1/10/2023	1/15/2023	Standard Class	AA-10780	Anthony Jacobs	Corporate	United States	Springfield
20	US-2023-149035	1/10/2023	1/15/2023	Standard Class	AA-10780	Anthony Jacobs	Corporate	United States	Springfield

Extract Table Using Examples Load Transform Data Cancel

Dataset preview displayed before the file is loaded into Power BI.



Superstore Sales data available in the Power BI report workspace.

Examining the Data in Power Query

Power Query Editor was opened to inspect the imported table in more detail. The columns, values, data types, and data-quality indicators were reviewed. The Applied Steps area also provides a record of the transformations already carried out, helping to track the preparation workflow.

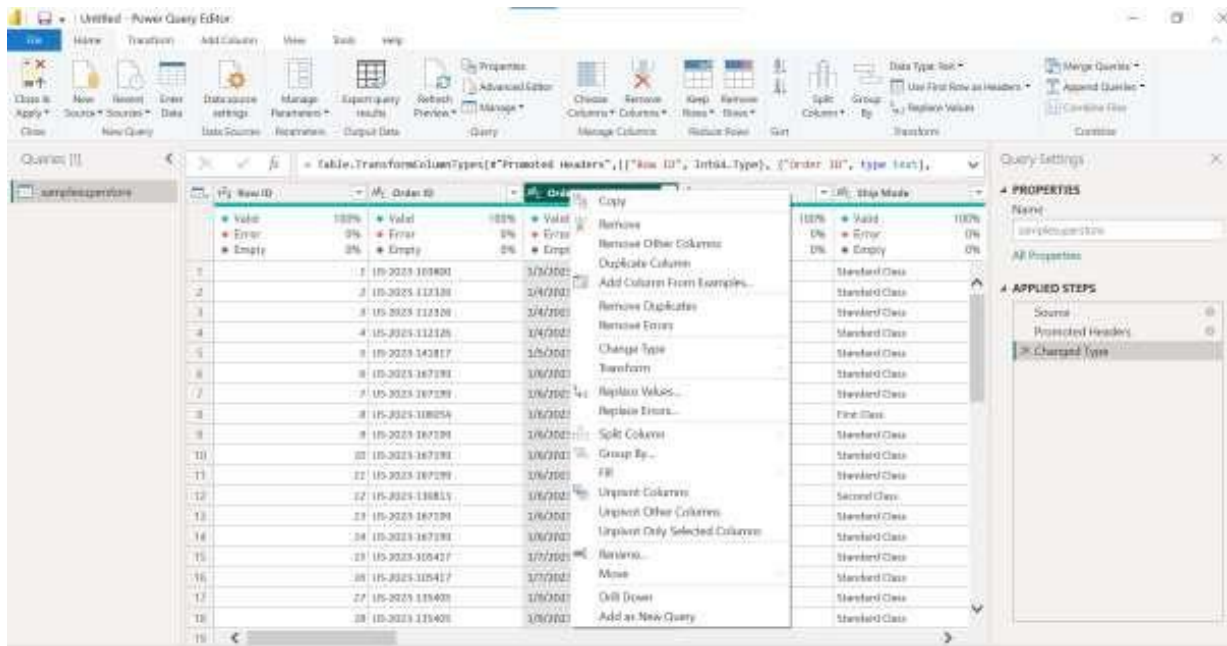
The screenshot displays the Power Query Editor interface. The main area shows a table with the following columns: Row ID, Order ID, Order Date, Ship Date, and Ship Mode. The table contains 18 rows of data. The right-hand pane shows the 'APPLIED STEPS' list with the following steps: Source, Promoted Headers, and Changed Type. The 'Changed Type' step is currently selected.

Row ID	Order ID	Order Date	Ship Date	Ship Mode
1	US-2023-103800	1/3/2023	1/11/2023	Standard Class
2	US-2023-112300	1/4/2023	1/6/2023	Standard Class
3	US-2023-112306	1/4/2023	1/6/2023	Standard Class
4	US-2023-112306	1/4/2023	1/6/2023	Standard Class
5	US-2023-141853	1/5/2023	1/12/2023	Standard Class
6	US-2023-147199	1/6/2023	1/10/2023	Standard Class
7	US-2023-147199	1/6/2023	1/10/2023	Standard Class
8	US-2023-147199	1/6/2023	1/10/2023	Standard Class
9	US-2023-147199	1/6/2023	1/10/2023	Standard Class
10	US-2023-147199	1/6/2023	1/10/2023	Standard Class
11	US-2023-147199	1/6/2023	1/10/2023	Standard Class
12	US-2023-147199	1/6/2023	1/10/2023	Standard Class
13	US-2023-147199	1/6/2023	1/10/2023	Standard Class
14	US-2023-147199	1/6/2023	1/10/2023	Standard Class
15	US-2023-147199	1/6/2023	1/10/2023	Standard Class
16	US-2023-147199	1/6/2023	1/10/2023	Standard Class
17	US-2023-147199	1/6/2023	1/10/2023	Standard Class
18	US-2023-147199	1/6/2023	1/10/2023	Standard Class

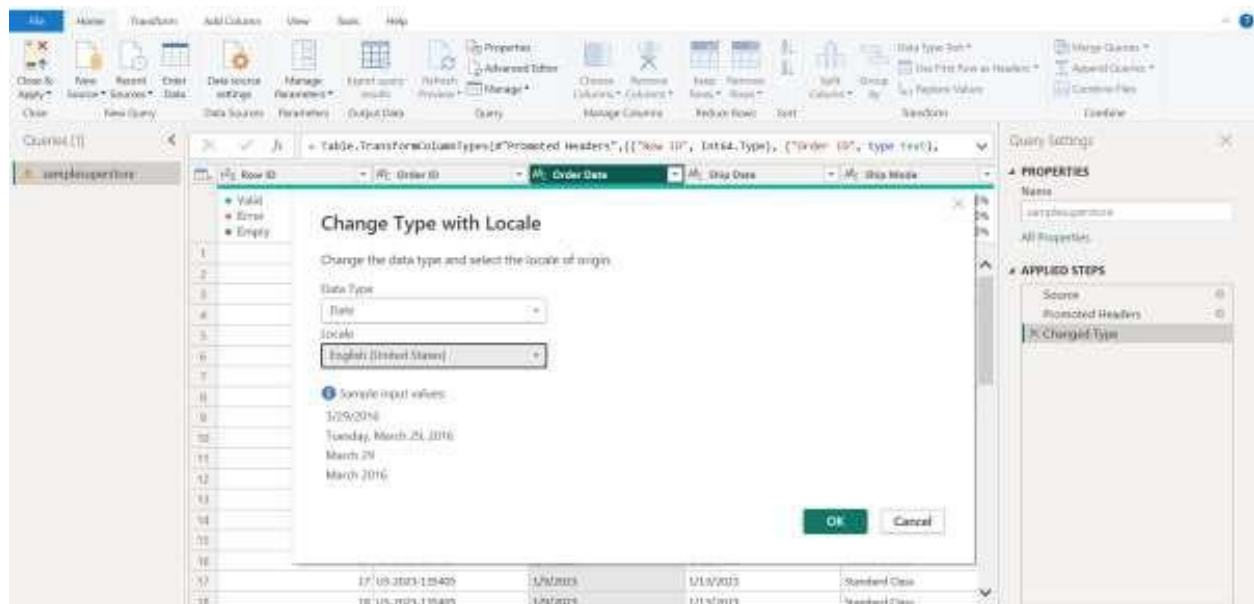
Superstore table inspected in Power Query Editor.

Preparing the Order Date Field

The Order Date column was prepared for reliable date-based analysis. The column was assigned a Date type and the English (United States) locale was selected so that the existing date values would be interpreted consistently. This makes the field suitable for later time-based reporting.



Change Type with Locale dialog used for the Order Date field.



Order Date values shown after conversion to a consistent date format.

Exploring Column Transformation Features

Power Query provides several commands for modifying an individual column. The transformation menu allows the user to change types, remove duplicate or error values, replace

entries, split fields, and rename columns. These options help shape the imported data into a cleaner analytical structure.

The screenshot shows a data transformation tool interface. At the top, there are tabs for 'Data Sources', 'Parameters', 'Output Data', 'Query', 'Manage Columns', 'Reduce Rows', 'Sort', and 'Transform'. Below these tabs, a formula bar contains the expression: `= Table.TransformColumnTypes(#"Changed Type", {"Order Date", type date}, "en-us")`. The main area displays a table with the following columns: Row ID, Order ID, Order Date, Ship Date, and Ship Mode. The 'Order Date' column is selected, and a transformation menu is displayed above it. The menu options are: Valid (100%), Error (0%), Empty (0%), and a dropdown for 'en-us'.

Row ID	Order ID	Order Date	Ship Date	Ship Mode
1	US-2023-103800	03-01-2023	1/7/2023	Standard Class
2	US-2023-112326	04-01-2023	1/8/2023	Standard Class
3	US-2023-112326	04-01-2023	1/8/2023	Standard Class
4	US-2023-112326	04-01-2023	1/8/2023	Standard Class
5	US-2023-141817	05-01-2023	1/12/2023	Standard Class
6	US-2023-167199	06-01-2023	1/10/2023	Standard Class
7	US-2023-167199	06-01-2023	1/10/2023	Standard Class
8	US-2023-108054	06-01-2023	1/7/2023	First Class
9	US-2023-167199	06-01-2023	1/10/2023	Standard Class
10	US-2023-167199	06-01-2023	1/10/2023	Standard Class
11	US-2023-167199	06-01-2023	1/10/2023	Standard Class
12	US-2023-130813	06-01-2023	1/8/2023	Second Class
13	US-2023-167199	06-01-2023	1/10/2023	Standard Class
14	US-2023-167199	06-01-2023	1/10/2023	Standard Class
15	US-2023-105417	07-01-2023	1/12/2023	Standard Class
16	US-2023-105417	07-01-2023	1/12/2023	Standard Class
17	US-2023-135405	09-01-2023	1/13/2023	Standard Class
18	US-2023-135405	09-01-2023	1/13/2023	Standard Class

Transformation menu displayed for the selected column.

Applying Date Type and Locale Settings

The selected field was again configured through the Change Type with Locale option. Choosing the Date data type together with the English (United States) locale supports consistent interpretation of date values and reduces formatting issues during subsequent analysis.

Data

settings

Parameters

results

Preview

Manage

Columns

Columns

Rows

Rows

Column

By

Replace Values

try

Data Sources

Parameters

Output Data

Query

Manage Columns

Reduce Rows

Sort

Transform

<

✕

✓

fx

= Table.TransformColumnTypes(*"Changed Type", ({"Order Date", type date}), "en-US")

▼

100%

Valid

0%

Error

0%

Empty

100%

Valid

0%

Error

0%

Empty

100%

Valid

0%

Error

0%

Empty

100%

Valid

0%

Error

0%

Empty

100%

Valid

0%

Error

0%

Empty

1

1

US-2023-103800

03-01-2023

1/7/2023

DP-13000

2

2

US-2023-112326

04-01-2023

1/8/2023

PO-19195

3

3

US-2023-112326

04-01-2023

1/8/2023

PO-19195

4

4

US-2023-112326

04-01-2023

1/8/2023

PO-19195

5

5

US-2023-141817

05-01-2023

1/12/2023

MB-18085

6

6

US-2023-167199

06-01-2023

1/10/2023

ME-17320

7

7

US-2023-167199

06-01-2023

1/10/2023

ME-17320

8

8

US-2023-106054

06-01-2023

1/7/2023

JO-15145

9

9

US-2023-167199

06-01-2023

1/10/2023

ME-17320

10

10

US-2023-167199

06-01-2023

1/10/2023

ME-17320

11

11

US-2023-167199

06-01-2023

1/10/2023

ME-17320

12

12

US-2023-130813

06-01-2023

1/8/2023

LS-17230

13

13

US-2023-167199

06-01-2023

1/10/2023

ME-17320

14

14

US-2023-167199

06-01-2023

1/10/2023

ME-17320

15

15

US-2023-105417

07-01-2023

1/12/2023

VS-21820

16

16

US-2023-105417

07-01-2023

1/12/2023

VS-21820

17

17

US-2023-135405

09-01-2023

1/13/2023

MS-17830

18

18

US-2023-135405

09-01-2023

1/13/2023

MS-17830

Copy

Remove

Remove Other Columns

Duplicate Column

Add Column From Examples...

Remove Duplicates

Remove Errors

Change Type

Transform

Replace Values...

Replace Errors...

Split Column

Group By...

Fill

Unpivot Columns

Unpivot Other Columns

Unpivot Only Selected Columns

Rename...

Move

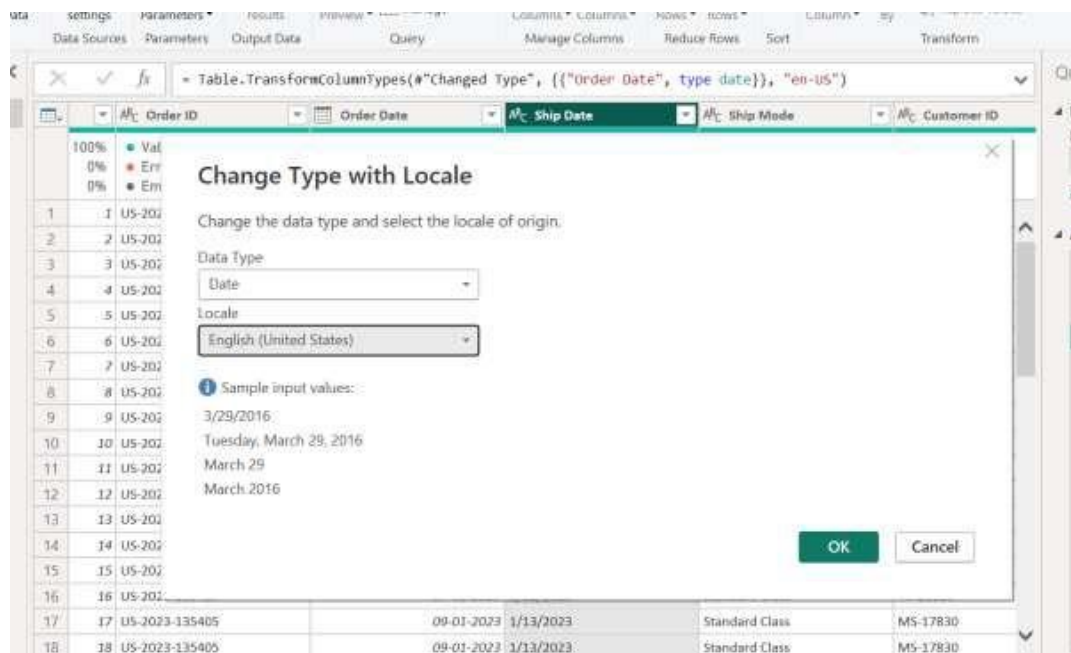
Drill Down

Add as New Query

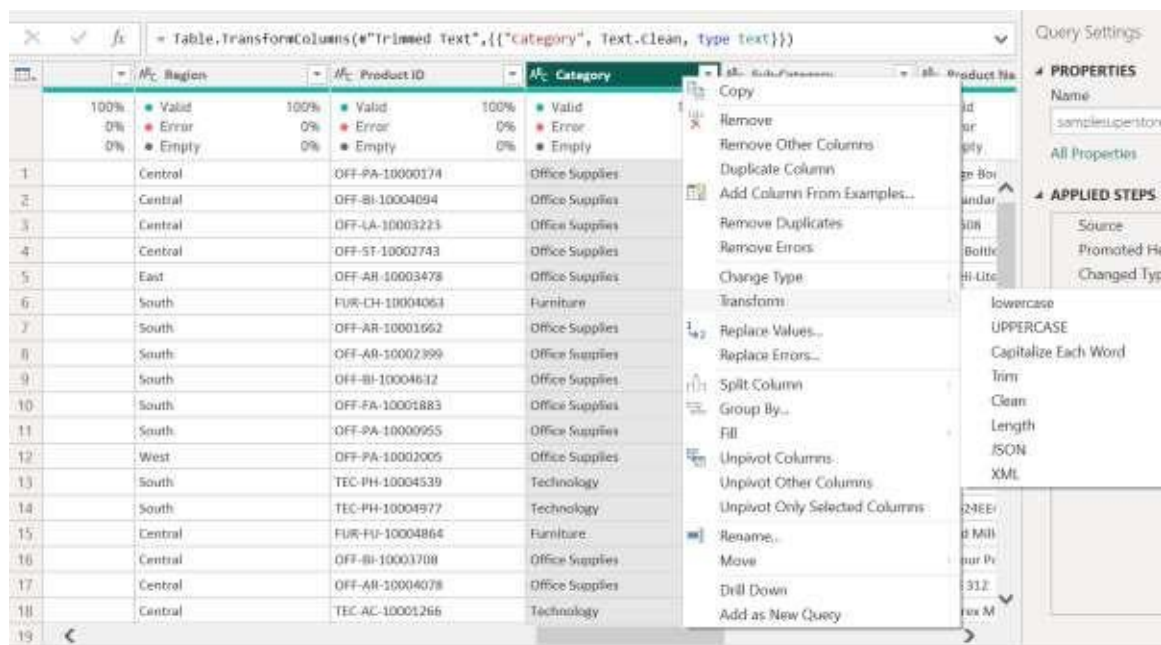
Date type and locale selection shown in Power Query.

Standardizing Categorical Text Fields

Text transformation commands were explored for categorical fields. Functions such as Trim, Clean, lowercase, UPPERCASE, and Capitalize Each Word can be used to make text values more uniform. Standardized category labels are easier to group, filter, and visualize in Power BI.



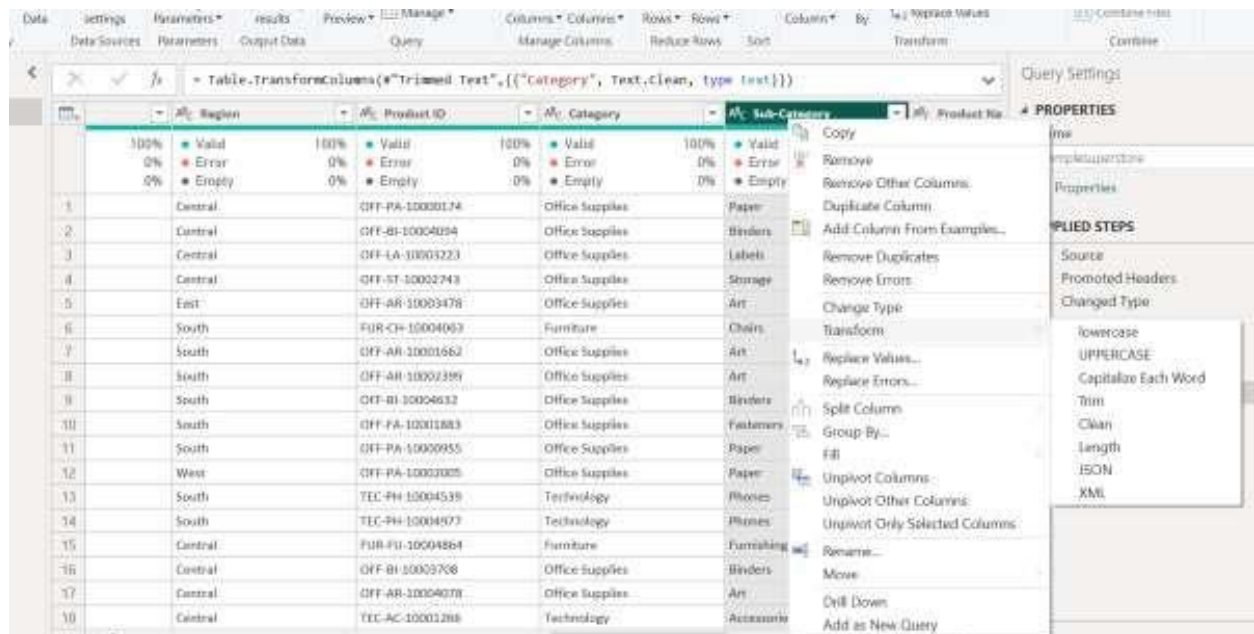
Text transformation commands available for the Category field.



Text transformation tools accessed for the Sub-Category field.

Cleaning the Sub-Category Values

The Sub-Category field was prepared using text-cleaning features. These operations help remove unnecessary spaces or unwanted characters and keep similar labels in a consistent form. This improves the reliability and readability of categorical analysis.



Cleaning and transformation options shown for Sub-Category.

Deriving Month Information from Order Date

A separate Month field was created from Order Date using the date options under Add Column. Creating this additional time attribute makes it easier to organize sales records into monthly periods and supports future trend analysis.

Table.TransformColumns(#"Trimmed Text1",{"Sub-Category", Text.Clean, type text})

APC Ship Mode

APC Customer ID

APC Customer Name

APC Segment

APC Customer/Region

100%

Valid

0%

Error

0%

Empty

100%

Valid

0%

Error

0%

Empty

100%

Valid

0%

Error

0%

Empty

100%

Valid

0%

Error

0%

Empty

1

07-01-2023

Standard Class

DP-13000

Darren Powers

Consumer

2

06-01-2023

Standard Class

PO-19195

Phillina Ober

Home Office

3

08-01-2023

Standard Class

PO-19195

Phillina Ober

Home Office

4

08-01-2023

Standard Class

PO-19195

Phillina Ober

Home Office

5

12-01-2023

Standard Class

MB-58085

Mick Brown

Consumer

6

10-01-2023

Standard Class

ME-17320

Maria Etezadi

Home Office

7

10-01-2023

Standard Class

ME-17320

Maria Etezadi

Home Office

8

07-01-2023

First Class

JO-15143

Jack O'Brian

Corporate

9

10-01-2023

Standard Class

ME-17320

Maria Etezadi

Home Office

10

10-01-2023

Standard Class

ME-17320

Maria Etezadi

Home Office

11

10-01-2023

Standard Class

ME-17320

Maria Etezadi

Home Office

12

08-01-2023

Second Class

LS-12230

Lynora Saunders

Consumer

13

10-01-2023

Standard Class

ME-17320

Maria Etezadi

Home Office

14

10-01-2023

Standard Class

ME-17320

Maria Etezadi

Home Office

15

12-01-2023

Standard Class

VS-21820

Viviek Sundaram

Consumer

16

12-01-2023

Standard Class

VS-21820

Viviek Sundaram

Consumer

17

13-01-2023

Standard Class

MS-17830

Melanie Seibe

Consumer

18

13-01-2023

Standard Class

MS-17830

Melanie Seibe

Consumer

Copy

Remove

Remove Other Columns

Duplicate Columns

Add Column From Examples...

Remove Duplicates

Remove Errors

Change Type

Transform

Replace Values...

Replace Errors...

Split Columns

Group By...

Pivot

Unpivot Columns

Unpivot Other Columns

Unpivot Only Selected Columns

Rename...

Move

Drill Down

Add as New Query

Query Settings

PROPERTIES

APPLIED STEPS

Source

Promoted Headers

Changed Type

lowercase

UPPERCASE

Capitalize Each Word

Trim

Clean

Length

JSON

XML

Month field being derived from the Order Date column.

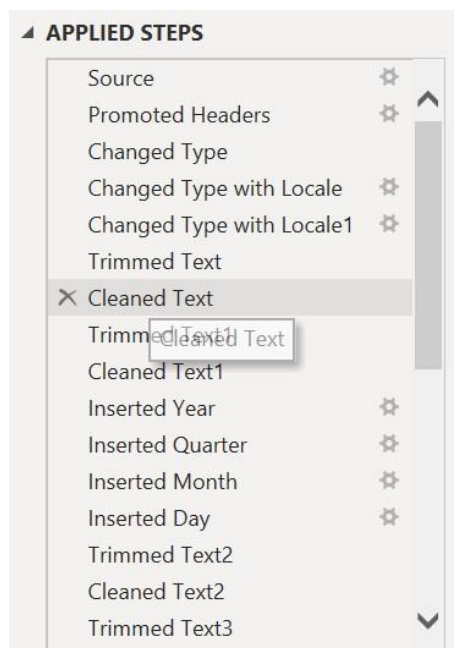
Base ID	Order ID	Order Date
133	133	115-2023-104280
134	134	115-2023-110200
135	135	115-2023-108312
136	136	115-2023-113000
137	137	115-2023-111880
138	138	115-2023-113000
139	139	115-2023-108412
140	140	115-2023-143302
141	141	115-2023-107324
142	142	115-2023-111157
143	143	115-2023-107324
144	144	115-2023-111157
145	145	115-2023-117935
146	146	115-2023-108648
147	147	115-2023-110425
148	148	115-2023-155852
149	149	115-2023-117944
150	150	115-2023-117944

Power Query date options used while creating the new Month field.

Tracking the Data Preparation Workflow

The Applied Steps panel records the sequence of data preparation actions. The workflow includes cleaning text values and deriving Year, Quarter, Month, and Day information from the

date field. Keeping these steps visible makes the transformation easier to process review and reproduce.



Applied Steps list showing the sequence of cleaning and date transformations.

Completing the Cleaning Process and Summary

The data preparation continued across the required text columns using trimming and cleaning operations. The final preparation stage leaves the Superstore Sales data organized for later reporting and visualization. Overall, the dataset was imported successfully, important fields were assigned appropriate types, text values were standardized, and useful date attributes were derived. The recorded Applied Steps provide a clear audit trail of the preparation process.



Additional Applied Steps showing continued cleaning of text fields.